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Abstract

A quantitative understanding of particle rigid body (RB) motion that inherently accompanies grain boundary (GB) diffusion is highly desirable to understand and control the dynamic interplay between coarsening and densification during solid state sintering. By computer simulation using a multi-phase-field approach, we analyze systematically the roles played by each of these processes at different stages of the shrinkage of the internal pore in a three-particle green body as a function of particle size as well as thermodynamic and kinetic factors of interfaces. We demonstrate that particle RB translation promotes both neck growth, and pore rounding and shrinkage. Moreover, the forces acting at GBs and pulling neighboring particles towards one another dynamically evolve as particles fuse. In contrast, particle RB rotation has no contribution to pore shrinkage. The translation force acting on an individual particle varies with not only its size, but also the number and sizes of its neighboring particles.

Keywords: sintering, phase-field modeling, microstructure stability, coarsening, densification

1. Introduction

Sintering is a processing technique for producing density-controlled materials and components from metal or ceramic powders by applying thermal energy [1]. Although practiced for over 28,000 years, sintering has continually found new applications [2-4]. For example, new energy systems, such as solar cells [5] and nuclear reactors[6], are critically contingent on sintered structures. Other examples include the fabrication of dental crowns and bridges using additive computer-driven sintering routes [7], and the production of porous tissue scaffolds for biomedical implants using custom laser sintering[8]. Most recently, there has been considerable interest and effort in pushing forward the sintering of thin printed electronic structures, thermoelectric junctions and long-lasting oil and gas drilling tools (e.g., bonded sintered diamond and cemented carbides[9]). Finally, sintering is a key step in the development of ceramic-based solid-state electrolytes that are critical for new batteries with improved energy efficiency and safety [10, 11]. There is no doubt that the applications of sintering in the fabrication of advanced materials for both structural and functional applications will continue to expand for many years to come [12]. Consequently, a fundamental understanding of sintering mechanisms and the processing/microstructure relationship is highly desirable for engineering desired microstructures for optimal properties targeting specific applications.

Sintering is driven by the reduction of surface free energy of a consolidated mass of particles and results in a redistribution of material in the contact surfaces between powder particles [1, 13, 14]. During sintering, the microstructure progression goes from particles in contact to a dense, nearly pore-free compact. The bonding process dynamically interacts with the continuously evolving microstructure via several concurrent mass transport processes. Transport mechanisms detail the paths by which the atoms move upon sintering. Since pores are large accumulations of vacancies, transport mechanisms describe vacancy migration and annihilation. Specifically, vacancies and atoms move along particle surface (i.e., surface diffusion), across pores (i.e., vapor transport through evaporation and condensation), along grain boundaries (GBs) (i.e., GB diffusion), and through the lattice (volume diffusion), as is schematically illustrated in Fig. 1(a) for a system of three particles during sintering [1]. In addition, vacancies may couple with dislocations via plastic flow and dislocation climb.

Transport mechanisms during sintering can be classified into two categories [1, 13, 15]: surface transport and bulk transport. The distinction is made based on whether or not the mass transport leads to densification. Surface transport mechanisms lead to neck growth without a change in particle spacing (i.e., no densification, Fig. 1(b)) and thus are non-densifying in that the mass flow originates and terminates at the particle surface. In other words, the atoms are rearranged with no annihilation of vacancies involved. Vapor transport, surface diffusion and lattice diffusion from the particle surfaces to the neck (see mechanisms 1, 2, and 3 in Fig. 1(a)) are the major contributors to surface transport-controlled sintering. By contrast, bulk transport mechanisms promote not only neck growth but also densification (see Fig. 1(c)). The paths involve vacancy flow from the neck surface to the interparticle GB, which produces densification since a layer of atoms moves in the opposite direction to the contact between the particles, thereby allowing the centers to approach each other as the sintering bond grows. GB diffusion and lattice diffusion from the GB to the pore (see mechanisms 4 and 5) are the most important densifying mechanisms in polycrystalline ceramics. Plastic flow by dislocation motion (see mechanism 6) also leads to neck growth and densification but is more common in the sintering of metal powders. Clearly, GBs are the source of materials transport for densification in crystalline powder compacts and GB diffusion appears to dominate the densification of many crystalline materials.

A cooperative GB accommodation of rigid-body motion is implied in the densifying path. Densification results from vacancy annihilation at GBs that act as a vacancy sink. The chemical potential of an atom is higher at GBs than at neck surfaces (which have higher negative curvatures), leading to a diffusional flux of atoms from GBs to nearby neck regions. This is equivalent to a vacancy flux directed to the GB regions under the assumption that vacancies are the only type of point defects present in the system. The over-saturated vacancies are then annihilated by dislocations at GBs, producing an effective force that pulls neighboring particles toward one another through rigid-body (RB) translation and rotation. The approach of neighboring particle centers through RB motion reduces interparticle spacing and thus plays an important role in the densification of powder compacts.

It is well known that sintering involves competition between the coarsening and densification of particles. Surface diffusion dominates at early stages of sintering and causes coarsening and neck growth without densification, whereas GB and bulk diffusion are more important at the intermediate stage or at high temperature, leading to densification. As such, quantification of the effect of particle RB motion accompanying GB diffusion and its influence on the interplay between coarsening and densification is a prerequisite to develop a comprehensive understanding of the physics underlying mass transport during sintering and to achieve better control of the sintering process. However, due to the dynamic interplay between the evolution of a heterogeneous microstructure (morphology and spatial distribution of solid grains and pores) and mass transport along the various paths mentioned above, it remains challenging to develop a quantitative understanding of the influence of particle RB motion by experimental study alone. A-physics-based sintering model must accurately capture all stages of sintering, as well as the transitions from one to another. This requires incorporating a variety of concurrent physical processes mentioned above. Nevertheless, most of the existing theoretical models for solid–state sintering subdivide the treatments into specific combinations of the sintering stages and mass transport mechanisms[16], such as surface diffusion during initial stage sintering or GB diffusion during intermediate stage sintering. Thus, these models are incapable of capturing the influence of particle RB motion at various stages of sintering.

Based on gradient thermodynamics, the phase-field approach (also called the diffuse-interface approach) offers an ideal framework to deal rigorously and realistically with these challenges [6, 7, 17-27]. In this approach, the total free energy (including the chemical, interfacial energy, and elastic strain energy if there is any) of an arbitrary heterogeneous system is formulated as a function of both the local chemical and structural states and their spatial variations (gradient)[28]. Microstructure evolution is driven by the reduction of total free energy through diffusion and structural relaxation and is described by the spatial and temporal evolution of the phase fields. One appealing feature of this method is that the various diffusion paths involved in sintering can be naturally taken into account by the dependence of the diffusivities on the microstructure features of the powder compact through phase-field variables without a need to explicitly track the interfaces, including surfaces and GBs[17]. The contributions from particle RB motion to mass flux can also be accounted for by introducing advection terms in both the Cahn–Hilliard nonlinear diffusion equation for conserved density fields and the time-dependent Ginzburg–Landau (i.e., Allen–Cahn) structural relaxation equation for non-conserved structural order parameters [17, 25, 29]. Several variants of phase-field model for solid-state sintering have been formulated and applied to study microstructure evolution and densification during different types of sintering. Insights from phase-field simulations of sintering pave the way for a better understanding of solid-state sintering [20-22, 25, 26], yet details about the role of particle RB translation and rotation at various stages of sintering and their individual and combined contributions compared with non-densifying mass transport mechanisms remain unclear.

Herein, by using a multi-phase-field model, we analyzed quantitatively, *for the first time*, the individual effects from particle RB motion (including translation and rotation) on microstructure evolution and densification kinetics at various stages of sintering as a function of particle size. It is found that particle RB translation leads to a faster densification (by promoting neck formation while preserving driving force for further neck growth and pore(s) elimination). In contrast, RB rotation has no influence on densification of spherical powder particles. RB translation dominates at the beginning and the initial intermediate stages of sintering. Smaller particles enable faster sintering due to enhanced RB particle motion.

The rest of paper is organized as follows: In Section 2, a three-dimensional (3D) multi-phase-field model is formulated. In Section 3, the model is employed to investigate the details of the entire sintering process and to identify the relative contributions from particle RB motion (translation and rotation) to microstructure evolution and densification at various stages of the sintering, as well as their dependence on powder size. In Section 4, the roles played by each of these mass transport mechanisms and their interplay at different stages during sintering are analyzed. The main findings are summarized in Section 5.

2. Methods

2.1. Multi-phase-field model for sintering

A multi-phase-field (MPF) model is formulated to gain a better understanding of the mass transport and microstructure evolution during sintering. As a generalized approach to treat multiple ($N > 2$) coexisting phases, the multi-phase-field (MPF) method is capable of treating an arbitrary number of individual phases that interact at triple or higher order junctions [28, 30-33]. An arbitrary sintering microstructure consists of a mixture of grains of solid phase and pores of vapor phase in a powder pellet and can be characterized by a density field $\rho(\mathbf{r})$ representing the relative mass density, and $(N + 1)$ long-range order (LRO) parameter fields. In MPF framework, each solid particle is assumed to be a crystalline grain with an unique orientation and is assigned a unique LRO, $\phi_\alpha(\mathbf{r})$ $(\alpha = 1 \ldots N)$, where N denotes the total number of particles in a green body [17, 29, 34]. The spatial distribution of the surrounding vapor phase is described by a specific LRO namely $\phi_{\alpha=N+1}(\mathbf{r})$. The phase fields, $\{\phi_\alpha\}$, define local volume fractions of all the phases present and, by definition, follow an additional constraint $\sum_{\alpha=1}^{N+1} \phi_\alpha(\mathbf{r}) = 1$ at any given grid point in the computational domain [28]. The bracket denotes the ensemble of all phases. $\mathbf{r}$ denotes the spatial coordinate.

The total free energy *functional* is an integral of the bulk chemical free energy density, f^{chem}, and the interfacial energy density, f^{intf}, over the system volume:

$$\mathcal{F}(\rho, \{\phi_\alpha\}) = \int d\mathbf{r} \left[f^{\text{intf}}(\{\phi_\alpha\}) + f^{\text{chem}}(\rho, \{\phi_\alpha\}) \right] \tag{1}$$

The bulk chemical free energy density is given by

$$f^{\text{chem}}(\rho, \{\phi_\alpha\}) = \sum_{\alpha=1}^{N+1} \phi_\alpha(\mathbf{r}) f_\alpha[\rho(\mathbf{r})] \tag{2}$$

where $f_\alpha[\rho(\mathbf{r})\,]$represents the free-energy density of the α^{th} phases given by

$$f_\alpha[\rho(\mathbf{x})] = \Delta f_0 V_m^{-1} [\rho - \rho_\alpha^0]^2 \tag{3}$$

with $\rho_\alpha^0 = 1(\alpha = 1 \ldots N)$ and $\rho_{\alpha=N+1}^0 = 0$ for the solid and vapor phases, respectively. Note that the free energy density $f^{\text{chem}}(\rho_\alpha, \{\phi_\alpha\})$ has a minima at $\left(\rho_{\alpha=1\ldots N} = 1, \phi_\alpha = 1,\ \phi_{\beta\neq\alpha} = 0\right)$ and $\left(\rho_{\alpha=N+1} = 0, \phi_{\alpha=N+1} = 1,\ \phi_{\beta\neq\alpha} = 0\right)$ for the solid and vapor phases, respectively. Such a free energy density provides an equilibrium state corresponding to a solid phase of crystalline grains of different crystallographic orientations and a vapor phase of both internal pores and the surrounding atmosphere. Δf_0 is a constant and V_m is the molar volume (assumed to be a constant). Based on the gradient thermodynamics [35], the interfacial energy density is defined by the gradients of the phase-fields and the potential functions:

$$f^{\text{intf}}(\{\phi_\alpha\}) = \sum_{\alpha=1}^{N+1} \sum_{\beta>\alpha}^{N+1} \left[-\frac{\kappa_{\alpha\beta}}{2} \nabla\phi_\alpha \cdot \nabla\phi_\beta + \omega_{\alpha\beta} \left|\phi_\alpha \phi_\beta\right| \right] \tag{4}$$

where $\kappa_{\alpha\beta}$ and $\omega_{\alpha\beta}$ are the gradient energy coefficients of and potential energy humps across the interface between the α and β phases, and they together determine the interfacial energy $\sigma_{\alpha\beta}$ and boundary width $\Lambda_{\alpha\beta}$.

The temporal and spatial evolution of the density and phase fields are governed by Cahn-Hilliard generalized diffusion equation and the time-dependent Ginzburg-Landau equation (Allen-Cahn), respectively, both of which are further modified by introducing an advection term accounting for the contribution from particle rigid-body motion. Such a treatment essentially separates the mass flux into two distinct contributions from diffusion[17], $\mathbf{J}^{\mathbf{diff}}$, and rigid-body motion, $\mathbf{J}^{\mathbf{adv}}$, in the continuity equation as follows:

$$\frac{\partial \rho}{\partial \mathrm{t}} = -\nabla \cdot (\mathbf{J}^{\mathbf{dif}} + \mathbf{J}^{\mathbf{adv}}) = \nabla \cdot \left(D(\phi, \rho) \nabla \frac{\delta F}{\delta \rho} \right) - \nabla \cdot \left(\rho \, \mathbf{v}^{\mathbf{adv}} \right) \tag{5}$$

$$\frac{\partial \phi_\alpha}{\partial \mathrm{t}} = -\sum_{\beta=1}^{\tilde{N}} \frac{L_{\alpha\beta}}{\tilde{N}} \left(\frac{\delta F}{\delta \phi_\alpha} - \frac{\delta F}{\delta \rho} \right) - \nabla \cdot \left(\phi_\alpha \, \mathbf{v}_\alpha^{\mathrm{adv}} \right) \tag{6}$$

In Eq. (5), $D\ (\phi, \rho)$ denotes the diffusivity and is a function of both ϕ_α and ρ [17, 34]:

$$D = h(\rho) D_{\mathrm{Bulk}} + [1 - h(\rho)] D_{\mathrm{Vap}} + 4\rho(1-\rho) D_{\mathrm{S}} + 4 D_{\mathrm{GB}} \sum_{\alpha}^{N} \sum_{\beta \neq \alpha}^{N} \phi_\alpha \phi_\beta \tag{7}$$

where $h(\rho)$ is an interpolation function [17],

$$h(\rho) = \rho^3 (10 - 15\rho + 6\rho^2) \tag{8}$$

which is used to distinguish the bulk solid phase from the vapor phase (a shape function), i.e., $h(\rho) = 0$ for the vapor phase and $h(\rho) = 1$ for the solid phase. D_{Bulk}, D_{Vap}, D_{GB}, and D_{S} characterize diffusivities in the solid bulk, in the vapor, along GBs and along the free surface, respectively. Note that Eq. (8) accounts for various diffusion paths involved during sintering (i.e., along surface and GB, through bulk lattice and vapor) without a need to explicitly track the locations of interfaces, including GBs and powder surfaces. $L_{\alpha\beta}$ is the phase-field mobility (which characterizes the kinetics of structural changes) and $\tilde{N}$ is the number of phases that locally coexist. Note that $\tilde{N}$ is different from N in Eq. (6), which denotes the total number of phases/grains in the system.

2.2. Advective flux due to particle rigid-body (RB) motion

The advective flux, $\mathbf{J}^{\mathbf{adv}} \equiv \rho \, \mathbf{v}^{\mathbf{adv}}$, accounts for mass transport arising from RB motion of a local volume element. Since RB motion consists of both translation and rotation, the advection velocity field, $\mathbf{v}^{\mathbf{adv}}$, is treated as a sum of the two contribution from all particles involved in the powder compact [17, 34]:

$$\mathbf{v}^{\mathbf{adv}} = \sum_{\alpha} \mathbf{v}_\alpha^{\mathrm{adv}} = \sum_{\boldsymbol{\alpha}} [\mathbf{v}_t(\mathbf{r}, \alpha) + \mathbf{v}_r(\mathbf{r}, \alpha)] \tag{9}$$

where $v_t(\mathbf{r}, \alpha)$ and $v_r(\mathbf{r}, \alpha)$ represent the velocity fields associated with the translation and rotation of the α^{th} particle which have been respectively formulated as follows:

$$\mathbf{v}_t(\mathbf{r}, \alpha) = \frac{K_t}{V(\alpha)} \mathbf{F}(\alpha) \phi(\mathbf{r}, \alpha) \tag{10}$$

$$\mathbf{v}_r(\mathbf{r}, \alpha) = \frac{K_r}{V(\alpha)} \mathbf{T}(\alpha) \times [\mathbf{r} - \mathbf{r}_c(\alpha)] \phi(\mathbf{r}, \alpha) \tag{11}$$

where K_t and K_r are the mobilities of the particle translational and rotational movement, respectively. $V(\alpha) \equiv \int \phi(\mathbf{r}, \alpha) \mathrm{d}^3\mathbf{r}$ defines the volume of the α^{th} particle, and $\mathbf{r}_c(\alpha) \equiv \int \mathbf{r} \cdot \phi(\mathbf{r}, \alpha) \mathrm{d}^3\mathbf{r}/V(\alpha)$ represents the center of mass of the α^{th} particle. It is worth noting that both force-velocity formulae of Eqs. (10) and (11) assume an inverse proportionality to the particle volume to ensure a law of inertia for the whole powder compact, i.e., the center of the mass of the powder compact does not change. This applies to free sintering (i.e., without constraint) as we consider in the current study.

$\mathbf{F}(\alpha)$ and $\mathbf{T}(\alpha)$ are the translational and torque forces acting on the α^{th} particle about its mass center and can be respectively formulated as:

$$\mathbf{F}(\mathbf{r}, \alpha) = \int \mathrm{d}\mathbf{F}(\mathbf{r}, \alpha) \tag{12}$$

$$\mathbf{T}(\mathbf{r}, \alpha) = \int [\mathbf{r} - \mathbf{r}_c(\mathbf{r}, \alpha)] \times \mathrm{d}\mathbf{F}(\mathbf{r}, \alpha) \tag{13}$$

$\mathrm{d}\mathbf{F}(\mathbf{r}, \alpha)$ represents the density of an effective local force acting on the GB of the α^{th} particle, which can be formulated in terms of the phase field order parameters as follows: (the effective force at GBs is caused by the mass density change at GBs)

$$\mathrm{d}\mathbf{F}(\mathbf{r}, \alpha) = \kappa \sum_{\beta \neq \alpha} (\rho - \rho_{\mathrm{e}}) \langle \phi(\alpha) \phi(\beta) \rangle [\nabla\phi(\alpha) - \nabla\phi(\beta)] \mathrm{d}^3\mathbf{r} \tag{14}$$

where ρ_{e} is a constant characterizing the equilibrium value of mass density at a GB. Note that the force originates from the variation of mass density with respect to ρ_{e} and vanishes if $\rho = \rho_{\mathrm{e}}$. κ is a stiffness constant that correlates the force magnitude to the vacancy over-saturation at GBs that causes density variation.

Eq. (14) essentially serves as a mathematical device that connects neighboring grains with a layer of "elastic glue" that maneuvers to maintain an equilibrium mass density at GBs [17]. Consequently, the neighboring particles could be either pulled closer or pushed further apart according to the mass density variation at their mutual GBs. Eq. (14) has two key features: The first feature is the threshold operation $\langle \phi(\alpha) \phi(\beta) \rangle$ employed to identify the GB region between the α^{th} and β^{th} particles through LROs, which is formulated as:

$$\langle\phi(\alpha)\phi(\beta)\rangle = \begin{cases} 1, & \text{if } \phi(\alpha)\phi(\beta) \geq c, \\ 0, & \text{otherwise} \end{cases} \tag{15}$$

where c is a critical value in identifying GBs through the product between $\phi(\alpha)$ and $\phi(\beta)$. Such operation excludes the region around the neck surface where the value of $\phi(\alpha)\phi(\beta)$ does not completely vanish while $\rho(\mathbf{r})$ significantly drops below the equilibrium value. The second feature is the gradient operation, $\nabla\phi(\alpha) - \nabla\phi(\beta)$, that ensures an action-reaction law between any pair of neighboring particles indexed by α and β, (i.e., the local forces between neighboring particles indexed α and β are equal in magnitude but opposite in direction).

2.3. Simulation set-up

All simulations were performed using a system having a volume of $96\Delta\mathrm{x} \times 96\Delta\mathrm{y} \times 96\Delta\mathrm{z}$ with the spatial step size $\Delta\mathrm{x} = \Delta\mathrm{y} = \Delta\mathrm{z} = l_0 = 25\ nm$ unless mentioned otherwise. Zero flux boundary conditions were used along all three dimensions. Both GB and surface energies were assumed to be isotropic, and the energy of each GB between any two different particles was further assumed to be the same. The energies used in the current study for GBs between different particles and powder surfaces are 0.68 and 1.0 $\mathrm{J/m^2}$, respectively unless mentioned otherwise. For the interface between arbitrary α and β phase, the boundary width $\Lambda_{\alpha\beta} = 6l_0$. The gradient energy coefficients $\kappa_{\alpha\beta}$ and potential energy hump $\omega_{\alpha\beta}$ can be computed by $\kappa_{\alpha\beta} = 8\sigma_{\alpha\beta}\Lambda_{\alpha\beta}/\pi^2$ and $\omega_{\alpha\beta} = 4\,\sigma_{\alpha\beta}/\Lambda_{\alpha\beta}$, respectively. Other model parameters include: the reference chemical mobility in the simulation $D_{\mathrm{Bulk}} = 1.5 \times 10^{-20}\ \mathrm{mol^2/(J \cdot s \cdot m)}$, $\Delta f_0 = 5 \times 10^4\ \mathrm{J/mol}$ (Eq. (3)) and molar volume $V_m = 1 \times 10^{-5}\ \mathrm{m^3/mol}$. As motivated by several studies[17, 21, 34], we consider GB and surface diffusion to be the active mass transport mechanisms by setting $(D_{\mathrm{Vap}}, D_{\mathrm{Bulk}}, , D_{\mathrm{GB}}, D_{\mathrm{S}}) = (10^{-1}, 10^{0}, 10^{1}, 10^{2})D_{\mathrm{Bulk}}$. The governing equations (Eqs. (5) and (6)) are also nondimensionalized with respect to the computational grid size l_0 and energy barrier Δf. The values of reduced parameters used in the simulations are: $\widetilde{D}_{\mathrm{Bulk}} = 1.50 \times 10^{-2}$, $\tilde{\kappa}_{\alpha\beta} = 7.78 \times 10^{-2}$ and 5.32×10^{-2} for surface and grain boundary, respectively. $\tilde{\omega}_{\alpha\beta} = 1.00 \times 10^{-2}$ and 7.29×10^{-3} for surface and grain boundary, respectively. $\kappa = 1$, $\rho_{\mathrm{e}} = 0.9816$ and $c = 0.14$ in Eq. (14). $\widetilde{K}_t$ and $\widetilde{K}_r$ will be specified in the Results section. $\tilde{L}_{\alpha\beta}$ is set to 1.08 to guarantee a diffusion-controlled sintering process.

We first started from a green body that consists of three spherical particles that have point contacts with each other. Such a three-particle geometry is schematically shown in Fig. 2(c), where three equal-sized spherical particles are centered at the vertices of an equilateral triangle located in the YZ plane through the center of the 3D simulation box. The radius of each particle is characterized by $R_\alpha = R$ $(\alpha = 1 \ldots 3)$. Such a spatial arrangement of the three particles leads to an internal pore with three cusps (a starting triangular shape shaded in gray, Fig. 2(d)). The sintering kinetics can be monitored by tracking the temporal evolution of the internal pore shrinkage metric, which is defined by:

$$\text{Shrinkage} = 1 - \frac{A(t)}{A(0)} \tag{16}$$

where $A(0)$ and $A(t)$ denote the cross-sectional areas of the pore (projected on the YZ plane through the center of the simulation box) at time 0 and t, respectively.

3. Simulation results

In this section, the multi-phase-field model is employed to examine the individual physical processes (in particular, surface and bulk mass transport) and their interplay that affects sintering neck growth, pore shrinkage and densification kinetics. In view of the complex nature of the sintering process, we first examine the surface and mass transport processes separately and then explore the consequence of their interaction, as well as their powder size dependence.

3.1. Sintering without densification

We start by considering the sintering of three particles (with size $R = 0.45\ \mu m$) involving surface mass transport processes alone. Note that grain boundary diffusion also operates in this case, but the attendant particle RB motion is excluded. Figs. 3(a)-(d) (top row) show the morphological evolution of the three particles and the internal pore in the middle of them during sintering. To better understand the mass transport along each path above upon sintering, the spatial distribution and temporal evolution of the diffusion flux $\mathbf{J}^{\mathbf{diff}}(\mathbf{r})$ in 3D are shown in the second row (Figs. 3(e)-(h)), in which the diffusion fluxes are visualized as red cones with their size proportional to the magnitude of the flux. The 2D contours of the three particles (solid green lines at which the values of $\phi_i = 0.5$, $i = 1 \dots 3$) are also superimposed to indicate the location of the surface and GBs (Figs. 3(e)-(h)). Moreover, the diffusion flux is also shown on 2D cross sections through the powder compact in the third row (Figs. 3(i)-(l)) in the YZ plane and in the bottom row (Figs. 3(m)-(p)) in the XZ plane (both planes section through the center of the simulation box), respectively. In the YZ plane, the sintering necks between any two neighboring particles can be clearly seen, while in the XZ plane, only the sintering neck formed between particle 1 (in green) and particle 2 (red) is visible (see the upper particle in Figs. 3(m)-(p)). Also shown on two bisection planes are the corresponding contours of the projected density fields ($\rho = 0.5$).

Sintering initiates at the point contacts between particles (Fig. 2(c)). The early stage of sintering is characterized by the formation of sinter necks that bond these particles. The sinter necks are saddle surfaces due to the presence of a mixture of both concave and convex curvatures (i.e., the sinter neck is concave (curved inward), while the particle surface is convex (curved outward)). Spatial distribution of the diffusion flux indicates that mass transport is most active near the powder surfaces close to sintering necks, followed by along the GBs, in agreement with the findings from classical sintering theory as well as molecular dynamics simulations of sintering processes[36]. In particular, diffusional flux is directed towards the concave regions (i.e., necks, Fig.3(i)), leading to the formation of sintering necks with material transported from the nearby convex particle surface. This occurs because the chemical potential of atoms below a convex surface is higher than that of atoms below a concave surface. In the meantime, a GB emerges at the contact plane (Fig. 3(e)). At this stage, the neck size is so small that each neck grows independently. As an example, the neck between particles 1 and 2 (the upper circle in Fig. 3(m)) is still physically separated from particle 3 (i.e., the circle on the bottom).

As sintering progresses, neighboring necks continue to enlarge and start to overlap with each other. The GB area in the neck increases as the neighboring necks impinge (Fig. 3(f)). As can be seen from Fig. 3(n), the contour lines ($\rho = 0.5$) at the bottom of the upper circle and the top of the lower one become elongated towards each other and eventually merge to bind both circles (see Fig. 3(o)). During this stage, the pore is still open but shrinks continuously. This occurs because a diffusion flux exists from GBs to the nearby pore where the three necks meet (Figs. 3(j) and (k)), and GBs act as fast diffusion pathways. As a result, the region where the GB emerges at a surface becomes smoother as well. By the final stage of sintering, the spherical pore continues to shrink and eventually disappears, and a complete closure of the internal pore in the YZ plane is reached. (see Figs. 3(d), 3(h) and 3(l)).

The simulation results in Fig. 3 clearly show that sintering involves a dynamic interplay between microstructure evolution and mass transport processes. Coupled with the mass transport processes are the various geometric stages (e.g., neck growth, pore rounding and shrinkage) traversed during sintering. The spatial variation of curvature on the powder surface gives a chemical potential gradient that directs mass flow into the sintering bond. The deposited atoms change the neck size and shape. In turn, the concomitant reduction in curvature and surface area influences the flux, as can be seen from the reduced the magnitude of the flux vector as sintering proceeds (Fig. 3(i)-(l)).

The temporal evolution of the pore shrinkage metric experiences two major changes in slope before reaching a complete closure of the internal pore (Fig. 4(a)). The changes in slope indicate that the simulation captures three different regimes and the transition from one to another during the sintering process. In general, there is no clear distinction between sintering stages since the geometric progression varies from point to point in the microstructure. However, based on Fig. 4(a), the whole pore shrinkage during sintering can be divided into three overlapping stages: initial, intermediate, and final. The initial stage is characterized by the formation of necks between particles (see Figs. 4(b)-(f)). In the intermediate stage, neighboring grains impinge to round the pore in the middle, leading to progressive smoothing of its surface. The final stage involves the closure of the internal pore. As will be further discussed in Section 4.3, each regime is governed by different physical processes: The initial stage is dominated by surface relaxation, which is then taken over by GB spreading in the intermediate stage. Moreover, the reduced magnitude of slope over the three consecutive stages indicates increasingly slower kinetics as sintering proceeds.

It should be noted here that the closure of the internal pore is not caused by the densification between particles wherein the inter-particle spacing is reduced. Instead, it is caused by the growth of the sintering necks (i.e., GBs) between any two neighboring particles, which leads to the elimination of the internal pore. To allow for pore shrinkage through particle densification, particle RB motion should be considered as presented in the section below.

3.2. Sintering with densification through rigid-body motion of particles

In addition to shape change by non-densifying diffusion, each particle may move as a rigid body associated with densifying mass transport through GB diffusion. Since particle RB motion consists of both translation and rotation, their individual contributions to microstructure evolution and densification kinetics are investigated separately in this section.

3.2.1. The role of particle translational motion

Figure 5 presents the results of particle translational motion. The 3D morphological evolution of three particles and the internal pores upon reaching a complete closure of the internal pore (within $t = 0 - 3\ s$) are presented in the first column. Neck growth leads to an increase in the area of GBs between particle (and contact angle (i.e., dihedral angle 2ϕ) at the triple-phase junctions as well as the shrinkage of the internal pore, as can be seen from Figs. 5(a), (d) and (g). To analyze the influence of particle translational motion, the spatial distribution and temporal evolution of the attendant advective flux in both 3D and 2D (XY cross-section) are presented in the second and third column, respectively. The direction of advective fluxes within each particle indicate that particle translational motion would lead the centers of each particle to approach each other. Besides diffusional flux toward the internal pore through surface and GBs as mentioned in Section 3.1, advective flux also moves mass inside the solid and deposits it in the pore (i.e., the flux bring mass to fill the internal pore), thereby accelerating the shrinkage of the internal pore. As shown in Figs. 5(c), (f) and (i), owing to the substantial neck growth assisted by particle translational motion towards each other, the internal pore quickly becomes spherical and closed within first three seconds, along with the significant reduction in surface curvature as pointed out by the arrows. It is also found that the magnitude of advective flux reduces as sintering proceeds, as indicated by the reduced sizes of both arrows (2D) and cones (3D) over time. Thus, similar to non-densifying mass transport, the contribution from particle translational motion varies with sintering stages and reduces as sintering progresses.

To further explore the influence of particle translation, the kinetics of pore shrinkage for different $\widetilde{K}_t$s are compared in Fig. 6(a). $\widetilde{K}_t$ characterizes the mobility of particle translational motion ($\widetilde{K}_t = 0$ indicates the absence of particle translation). Apparently, consideration of particle translational motion ($\widetilde{K}_t > 0$) shortens the time needed for a full collapse of the internal pore under otherwise identical conditions. In particular, t_f decreases as the $\widetilde{K}_t$ increases to a value of 0.5 but is relatively unchanged at larger $\widetilde{K}_t$s (Fig. 6(b)). The plateau region indicates that the closure of the internal pore requires other cooperative physical process(es) with particle translation as particles fuse (see discussion at Section 4.2).

Besides pore shrinkage kinetics, particle translational motion is also found to affect the morphological evolution of the particles. Fig. 6(c) compares the 2D contour of the density ($\rho = 0.5$) for $\widetilde{K}_t = 0.5$ and $\widetilde{K}_t = 0.0$. At the selected time moment ($t = 3\ s$), the internal pore has completely collapsed for $\widetilde{K}_t = 0.5$, while it still remains open for $\widetilde{K} = 0.0$. In addition, the triple junction for $\widetilde{K}_t = 0.5$ is less concave (i.e., larger 2ϕ) than for $\widetilde{K} = 0.0$. This comparison suggests that particle RB motion bring particles closer to each other and thus promotes sintering neck growth and the shrinkage of the internal pore.

In order to understand the varying magnitudes of advective flux during sintering (see the third column in Fig. 5), we analyzed the density of the translational forces (i.e., $\mathbf{F}(\mathbf{r}, \alpha)$ in Eq. (12) normalized by the volume of the α^{th} particle) that lead to particle translational motion. The result is shown in Fig. 6(e) for particle 3 at different $\widetilde{K}_t$s. Note that the forces acting on particles 1 and 2 have the same magnitude but different directions. It is seen that the force starts to develop (once the threshold value is reached and the GB can be identified, see Eq. (15)) and increases sharply until reaching a maximum, and then decays and eventually vanishes (which explains the reduced advective flux over time as shown in Figs. 5(c), (f) and (i)). For each $\widetilde{K}_t$, the maximum in the force density is reached at an early stage of sintering, as highlighted by an open circle (Fig. 6(f)). The results indicate that the maximum force magnitude increases with decreasing $\widetilde{K}_t$, but it takes a longer time for the force to reach the maximum for a smaller $\widetilde{K}_t$. This occurs because the velocity (i.e., the product of force and mobility as calculated by Eq. (10)) of particle translation that brings the mass density at GBs to reach the equilibrium value ρ_e is much faster for larger $\widetilde{K}_t$s (see Fig. 6(d)).

3.2.2. The role of particle RB rotation

In this section, the individual influence of particle RB rotation is considered by incorporating the advective flux corresponding to rotation in Eqs. (5) and (6). The kinetics of pore shrinkage for different $\widetilde{K}_r$s are shown in Fig. 7(a). $\widetilde{K}_r$ characterizes the mobility of particle rotational motion ($\widetilde{K}_r = 0$ in the absence of particle rotation). The densification kinetics appear to be independent of the choice of $\widetilde{K}_r$. To validate this finding, the influence of particle RB rotation on pore shrinkage kinetics is quantified by $t_f/t_f(\widetilde{K}_r = 0)$, where t_f and $t_f(\widetilde{K}_r = 0)$ denote the time to reach a complete pore closure with and without considering rotation, respectively. Particle RB rotation would either enhance (if $t_f/t_f(\widetilde{K}_r = 0) < 1$) or suppress (if $t_f/t_f(\widetilde{K}_r = 0) < 1$) the pore shrinkage. The results are shown in Fig. 7(b) for different values of $\widetilde{K}_r$, where the maximum deviation of the value of $t_f/t_f(\widetilde{K}_r = 0)$ from 1 is less than 2.5%, indicating that particle rotation does not contribute to the shrinkage kinetics of the internal pore.

To explore the underlying physics of this finding, advection flux in 3D corresponding to particle RB rotation at an intermediate time moment is visualized in Fig. 7(c). Additionally, the advective flux associated with rotation is projected on three different XZ planes at different locations along the Y-axis (as indicated by the three dashed lines in the YZ cross-section in Fig. 7(d)), which are shown from left to right Figs. 7(e-g). It can be seen that the advective flux enters and exits the neck region with the same magnitude. In other words, no net mass from the inside of the solid particle is deposited at the neck region or in the pores during particle RB rotation. As a result, particle rotation does not contribute to the shrinkage of the internal pore. Note that the advective flux due to particle rotation is normal to the YZ cross-section. Thus, the flux shown in the YZ cross-section is the diffusional flux.

3.3. The influence of particle size on pore closure kinetics

One of the paradigms of sintering theory is that a reduction in particle size enhances the sintering kinetics or reduces the sintering temperature. A great deal of independent experiments have demonstrated the strong particle size dependence of sintering kinetics. However, the physical origins of the observed trends remain incompletely understood in terms of the individual and combined contributions from non-densifying and densifying mass transport processes. The size dependence of sintering kinetics has been primarily ascribed to the increase in both the driving force (through changes in curvature) and kinetics (through decrease in the length of diffusion path) associated with reduction in particle size, which both lead to faster sintering. Even though it has been recognized that rigid-body motion plays a nontrivial role in the densification of sintered powder pellets [17, 34], the particle size dependence of this effect is not yet completely understood. In this section we demonstrate that beyond reproducing experimental trends, one advantage of the model is the ability to discern which physical parameters dominate the observed behavior. Along these lines, we systematically analyzed the individual roles of different mass transport mechanisms in sintering kinetics as a function of particle size. Considering that particle rotation has no influence on pore shrinkage kinetics (see section 3.2.2), only the contribution from particle translation is considered below.

First, the size dependence of the non-densifying mechanism is addressed. Fig. 8(a) shows the pore shrinkage kinetics for three different powder sizes when the surface transport mechanism alone is operating. It is clear that the magnitudes of the slopes for each stage of pore shrinkage increase with decreasing particle size. As a result, t_f decreases with reduced particle size.

It is noted that clusters of small particles reach final stage sintering first, while regions of large particles might still be in the initial stage. For particles of $R = 300\ nm$, the growth of sintering necks continues even after closure of the internal pore at $t = 2.0\ s$ (see the comparison of density profile at $t = 2.0$ and $10.0\ s$ in Fig. 8(b)), which indicates that at this moment the neck size has not yet reached a thermodynamic equilibrium. In contrast, for particles of $R = 600\ nm$, a complete closure of the internal pore is not achieved even at $t = 16.0\ s$ (see Fig. 8(d)). Instead, the pore shrinkage stagnates at a fraction of ~ 0.55 when the time reaches $t = 14\ s$. Without a densifying mass transport mechanism operating, pore closure can only be achieved if there exists neck growth. The stagnation of neck growth indicates that the neck size has reached a thermodynamic equilibrium (i.e., $X = 2R\sin\phi$, see Fig. 1). Further migration of the grain boundary would require an increase in its area, making the process energetically unfavorable. This is supported by the observation that a much longer sintering time (e.g., $t = 40\ s$) does not change either neck size or the dihedral angle at the triple junction line.

In particular, the diffusion flux at an intermediate time point ($t = 1.0\ s$) is shown in Figs. 8(c) and 8(f) for $R = 300\ nm$ and $600\ nm$, respectively. Note that at the selected time, for the smaller particles sintering has already reached its final stage, while for the larger ones sintering is still at its early stage (see Fig. 8(a)). Thus, comparing the sizes of the arrows, we conclude that small particles sinter with enhanced diffusional flux.

Second, Fig. 9(a) shows the pore shrinkage kinetics for three different powder sizes when non-densifying and densifying mass transport mechanisms operate simultaneously during sintering. Compared with Fig. 8(a), particle translation promotes pore closure for all three different particle sizes. Pore closure is completed within a shorter time for particles of size $R = 300$ and $450\ nm$. Additionally, pore closure can be achieved for $R = 600\ nm$ when particle RB translation operates during sintering. Fig. 9(b) demonstrates temporal evolution of the translational force magnitude acting on particle 3 for different sizes. The most significant variation of force occurs within $[0\sim 1]\ s$ (Fig. 9(c)), in which the maximum in the force magnitude for each particle size is highlighted as an open circle. Clearly, the maximum force magnitude increases with decreasing particle size. As a result, the neighboring smaller particles are pulled towards one another by the force with increased magnitude, leading to enhanced densification. This is also manifested by the increased magnitude of the advective flux (larger arrow size) for smaller particles (see Figs. 9(d) and (e) for a comparison of the advective flux for particles with two different sizes at a time moment indicated by the vertical dashed line in Fig. 9(c)).

According to Eq. (10), the velocity for particle rigid-body motion among particles is inversely proportional to the particle volume, rather than the particle size as for the diffusion flux. Our quantitative results clearly indicate that the advective flux is more sensitive to the particle size than the diffusion flux (see the difference in advective fluxes in Figs. 8c and 8f for particle sizes of $R = 300\ nm$ and $R = 600\ nm$, respectively).

Finally, it is noted that the initial area of the internal pore in the three-particle green body is, $A_0 = \sqrt{3}R^2 - \pi R^2/4$. Thus, A_0 is proportional to the square of the radius and decreases with decreasing particle radius. A smaller pore takes less time to be eliminated than a larger one.

In summary, there are three factors responsible for the observed dependence of sintering kinetics on particle size (i.e., smaller particle sizes promote faster sintering). First, the diffusive flux (driven by the spatial variation of curvature on the particle surface) is inversely proportional to the particle radius. Second, the advective flux or velocity is inversely proportional to the particle volume, resulting in a larger advective flux/velocity for a smaller particle (see Figs. 8c and 8d). Third, the initial pore size is directly proportional to the square of the particle size, indicating that a smaller pore takes less time to be eliminated than a larger one.

4. Discussion

Sintering involves multiple simultaneous physical processes, including diffusion along various paths and particle rigid-body motions (translations and rotation) that are associated with GB diffusion and result in densification. Coupled with these mass transport pathways are the various geometric stages that evolve during sintering, including neck growth and the concomitant reduction in curvature and surface area, as well as pore rounding, shrinkage and closure. The simulation results above clearly demonstrate that the interplay among these processes varies with sintering stage as well as with particle size. Depending on the relative contributions from these factors, the relationship between coarsening and densification can be determined. The phase-field model naturally incorporates all of the above-mentioned processes and factors during sintering within a unified formalism in a physically consistent way. Beyond reproducing the experimental trends, one advantage of the model is the ability to allow for a quantitative understanding of each of the processes at different stages of sintering.

4.1. The effect of surface diffusion and GB diffusion on pore shrinkage

We identify the individual roles of surface diffusion and GB diffusion in the process of pore closure by systematically varying both thermodynamic (e.g., surface and GB energies) and kinetic (surface and GB diffusivities) factors related to these two processes.

Figure 10(a) shows pore shrinkage kinetics for different surface diffusivities D_ss, which are varied from $5.0\ D_{\mathrm{GB}}$ to $50\ D_{\mathrm{GB}}$ (i.e., $50.0\ D_{\mathrm{Bulk}}$ to $500.0\ D_{\mathrm{Bulk}}$) while keeping all other parameters unchanged. The slopes (see Fig. 10(b) for the early stage) for each stage of pore shrinkage increase with increasing D_ss, thereby shortening t_f. Fig. 10(c) summarizes t_f for different D_ss, which shows that as D_{s} increases by a factor of 10 ($50.0\ D_{\mathrm{Bulk}}$ to $500.0\ D_{\mathrm{Bulk}}$), t_f reduces by a factor of 8.7 (from $15.1\ s$ to $1.7\ s$). It is also noted that t_f decreases sharply as D_{S} increases to $\sim 100 \times D_{\mathrm{Bulk}}$, but further reduction slows down at larger values of D_{s}s. This trend indicates that pore shrinkage kinetics are enhanced by fast surface diffusion and that surface diffusion also requires other cooperative processes to achieve complete pore closure.

Figure 11 (a) shows pore shrinkage kinetics for different GB diffusivities D_{GB}s, which are varied from $0.1\ D_{\mathrm{s}}$ to $3.0\ D_{\mathrm{s}}$ (i.e., $10.0\ D_{\mathrm{Bulk}}$ to $300.0\ D_{\mathrm{Bulk}}$) while keeping all other parameters unchanged. It can be seen that the slopes for the initial stage seem to be independent of D_{GB} (see Fig. 11(b)), indicating that this stage is essentially dominated by surface diffusion (see Fig. 10(b) for a comparison). As sintering proceeds to the intermediate stage, the differences in pore shrinkage kinetics start to develop, with larger slope magnitudes for increasing D_{GB}. Figure 11(c) summarizes t_f as a function of D_{GB} and shows that as D_{GB} increases by a factor of 30, t_f decreases by a factor of 1.7. Thus, neck growth kinetics are much more sensitive to surface diffusion (see Figs. 10(c) and 11(c) for a comparison).

Considering that sintering is driven by surface energy reduction and is accompanied by the formation of sintering necks and thus the increase of grain boundary areas between particles, we further examine the dependence of sintering kinetics on surface energy (σ_s) and grain boundary energy (σ_{GB}). Fig. 12(a) shows the pore shrinkage kinetics for different σ_s values, which are increased from $1.0 \times \sigma_{\mathrm{GB}}$ to $2.2 \times \sigma_{\mathrm{GB}}$ while keeping all other parameters unchanged. Similar to surface diffusivity, t_f first decreases sharply with increasing σ_{s} but further reduction is slowed down at larger values of σ_{s}s. In total, as σ_s increases by a factor of 1.67 (from $1.32\ \sigma_{\mathrm{GB}}$ to $2.20\ \sigma_{\mathrm{GB}}$), t_f decreases by a factor of 2.65 (from $9.32\ s$ to $3.52\ s$). Besides pore shrinkage kinetics, surface energy σ_{s} also dictates the morphology of pores and grains. At thermodynamic equilibrium, the forces must balance at the junction where the surfaces of the pores meet the GB, which leads to $\sigma_{\mathrm{GB}} = 2\sigma_{\mathrm{s}} \cos\phi$, where 2ϕ is the dihedral angle (see Fig. 1(b)). Clearly, ϕ becomes larger for larger σ_{s}, as shown in Fig. 12(c), wherein the contours of the density field are compared for two surface energies at $t = 10\ s$ (note that the internal pore has not yet reached a complete closure at this time point for $\sigma_{\mathrm{s}} = 1.0 \times \sigma_{\mathrm{GB}}$).

Fig. 13(a) shows the pore shrinkage kinetics for different GB energies, σ_{GB}, which are reduced from $1.00 \times \sigma_{\mathrm{s}}$ to $0.25 \times \sigma_{\mathrm{s}}$. All other parameters are kept the same except that σ_{s} is increased from 0.68 to $1.00\ \mathrm{J/m^2}$. The differences in pore shrinkage kinetics for different σ_{GB} values start to develop only when sintering progresses into the intermediate stage. The increase of σ_{s} leads to much faster closure of the internal pore as compared to the case in Fig. 12(c) wherein $\sigma_{\mathrm{GB}} = 1.0 \times \sigma_{\mathrm{s}} = 0.68\ \mathrm{J/m^2}$. In total, σ_{GB} increases by a factor of 4.0 (from $0.25\ \sigma_{\mathrm{GB}}$ to $1.00\ \sigma_{\mathrm{GB}}$), while t_f increases by a factor of 2.68 (from $4.50\ s$ to $12.05\ s$). Clearly, t_f is more sensitive to the increase of σ_{s} than to the reduction of σ_{GB}.

The parametric studies above make clear the variation of the dominant mechanisms with the stages of sintering. The initial stage of pore shrinkage during sintering is characterized by neck growth. On the one hand, the increasing thermodynamic driving force (i.e., surface energy) and kinetic mobility (i.e., surface diffusivity) promote neck growth. On the other hand, the actual neck volume in the initial stage is relatively small. As such, the thermodynamic and kinetic factors associated with grain boundaries have trivial influence on pore shrinkage kinetics at this stage. As the sintering progresses and the GB area in the necks increases while the surface area decreases, GB diffusion starts to overtake surface diffusion. In this stage, reducing GB energy and increasing GB diffusivity promote neck growth. However, GB diffusion requires cooperative surface diffusion, for example, to avoid building hillocks where the GB emerges at a surface.

4.2. *The effect of non-densifying and densifying mass transport mechanisms on pore shrinkage*

It is noticed that, in the absence of particle translation, a complete closure of the internal pore does not occur when the particle size is large enough, e.g., $R = 600\ nm$ in the current study (see Figs. 8(a) and (g)). In contrast, the presence of particle RB translation helps to close the internal pore under otherwise identical conditions. This occurs because the densification promotes neck growth and thus preserves a high capillary driving force for sintering. The difference in pore shrinkage behaviors emphasizes the importance in accounting for the contribution from RB motion when evaluating the competition between coarsening and densification during sintering.

It is also noted that the reduction in t_f seems to be independent of $\widetilde{K}_t$ (i.e., the mobility of particle translational motion) at larger values of $\widetilde{K}_t$ beyond 0.5. In order to evaluate the relative contribution from non-densifying and densifying mass transport to pore shrinkage kinetics, computational experiments are designed in which both surface and GB diffusivity are varied systematically while keeping all other parameters the same. Fig. 14(a) shows the pore shrinkage kinetics for three different D_{s} values in the case of $\widetilde{K}_t = 1.0$. Like the previous case in Section 4.1, increasing D_{s} increases the magnitude of the slope for each pore shrinkage stage, thereby reducing t_f. Thus, the finding about the influence of surface diffusivity D_{s} on pore shrinkage is still valid in the presence of particle RB motion. Note that the value of D_{s} does not change the magnitude of maximum translational force (Fig 14(c)). Instead, the reduced D_{s} delays the time it takes for the translational force to reach its maximum (Figs 14(c) and (d)). As a neck grows between two particles in contact, there is a concomitant variation in mass density at the GB that results in translational forces (see Eq. (14)) pulling neighboring particles closer to each other. Fast surface diffusion facilitates mass transport and thus particle translation. It is clear that surface diffusion and particle translation are complementary to each other in promoting neck growth. The former induces the variation of mass density across the GBs between neighboring particles, which in turn generates forces that pull these particles closer to each other. Also, the density of translational force becomes relatively small as the pore shrinkage moves into its second stage wherein GB diffusion becomes more important.

Fig. 14(e) shows pore shrinkage kinetics for three different GB diffusivities, D_{GB}, in the case of $\widetilde{K}_t = 1.0$. Similar to the absence of particle RB translation, pore shrinkage starts to be influenced by GB diffusion only when it moves into the intermediate stage (see Fig. 14(f)). Different from surface diffusion, GB diffusion affects neither the translational force nor its temporal evolution (Figs. 14(g) and (h)). This is the case for relatively large values of $\widetilde{K}_t$. For relatively small values of $\widetilde{K}_t$ (see Fig. 6(a) and (d) for $\widetilde{K}_t = 0.01 \sim 0.1$), however, the translational force can persist as the pore shrinkage goes into the intermediate stage wherein GB diffusion becomes operative. Therefore, a large mobility of particle translation (e.g., $\widetilde{K}_t = 1.0$ in the current study) may help to bypass the initial stage, thereby reducing the coarsening or surface diffusion of particles and preserving a high capillary driving force for the subsequent sintering.

4.3. The influence of particle size asymmetry

The green body considered in all the simulations above consists of three identical spherical particles in point contact with each other. A realistic green body contains a broad distribution of initial particle sizes and a particle may be in contact with other particles of different sizes. Using the three-particle green body as an example, we analyze how differences in particle size influence the RB motion between them. Along this line, the radius of particle 3 is modified to be $R_3 = 300$ and $600\ nm$, while the radii of particles 1 and 2 are kept unchanged ($R_1 = R_2 = 450\ nm$). Fig. 15(a) shows pore shrinkage kinetics for three different R_3 values. Reducing R_3 alone increases the magnitude of the slope for each stage of pore shrinkage. As a result, t_f decreases from $4.11\ s$ to $1.77\ s$ as R_3 is reduced from $600\ nm$ to $300\ nm$. Fig. 15 (b) shows the temporal evolution of the density of translational force acting on particle 3 with three different sizes. It can be clearly seen that the maximum of force density (indicated by open circles) is larger for smaller particles. In contrast, for the particles neighboring particle 3, the translational force (e.g. for particle 1 $R_1 = 450\ nm$) increases with the size of its neighboring particle 3 (Fig. 15(c)), which is manifested by the increased arrow size with increasing R_3 (see the comparison between Fig. 15(d) for $R_3 = 300$ and (e) for $R_3 = 600\ nm$).

The difference in R_3 also modifies the direction of advective flux in particles 1 and 2 (Fig. 15(d) and (e)). Thus, the value of R_3 affects both the magnitude and direction of the translational forces acting on its neighboring particles. This finding suggests that the green body characteristics (e.g., particle size and distribution) could be optimized to improve the sintering kinetics or reduce the sintering temperature, since particle size distribution determines the number and size of neighboring particles for each particle and thus the magnitude and direction of the translational force acting on it. The developed phase-field model is well suited for identifying the optimum distribution(s).

The influence of particle size asymmetry on the particle rotational motion is also investigated and the results are presented in Fig. 16. The radius of particle 3 is modified to be $R_3 = 300\ nm$ while keeping the radii of the other two particles the same. The mobility of the rotational motion $\widetilde{K}_r$ is then varied from 0.01 to 10.0 and the influence of the variation of $\widetilde{K}_r$s on pore shrinkage kinetics is summarized in Fig. 16(a). The deviations of the values $t_f/t_f(\widetilde{K}_r = 0)$ for different $\widetilde{K}_r$s from 1 fall within 2.5%-5%, indicating that for spherical particle aggregates, particle RB rotation has no influence on sintering kinetics, regardless of the particle size distribution.

4.4. Limitations of the model

All the simulations are performed for a green body that consists of three particles in point contact with each other, and a single internal pore surrounded by the particles. However, a realistic green body is likely to contain a large number of particles (and thus pores) with a broad distribution of sizes and shapes. As such, the current study has ignored the process of grain growth (driven by the reduction of total grain boundary energy), the pore-boundary interactions, and the pore-pore interactions, as well as their influence on the dynamic interplay between densification and coarsening.

Besides particle size, it should be noted that there are many other processing parameters that influence densification and coarsening, including packing configuration and density of the green body, temperature profile (such as heating rate, sintering temperature and time duration, cooling rate), and the presence of sintering agents. However, we expect that the essential physical mechanisms and the relationships identified in our simplified model will not qualitatively change. As an example, we consider the sintering behavior of a more realistic green body that contains 1024 particles (Fig. 17(a)) with the size distribution informed by experimental measurements (Fig. 17(b)). Fig. 17(c) shows the temporal evolution of the translational forces acting on three randomly selected particles. Particles 1 (Dark red) and 2 (Yellow) are located on the green body surface while particle 3 is embedded in the green body interior. The results indicate that the development of forces on these three particles follow the same trend as observed in Fig. 6(e) and 9(b). The differences lie in several fluctuations in the force magnitude in the downhill portions of the three curves. The fluctuations occur because in a realistic green body, a particle is in contact with more than two neighboring particles that experience both coarsening and RB translation and thus a grain boundary could be perturbated by grain growth and/or pore/GB interaction. A more detailed analysis of microstructure evolution (that involves concomitant densification, pore rounding, and grain growth as shown in Figs 17(d)-(f)) during sintering is forthcoming but is beyond the scope of this paper.

The densification of a sintering compact is accommodated by particles' rigid body motion, which are accounted for phenomenologically by introducing an advective mass flux with an adjustable parameter (i.e., mobility) that accounts for the time scale associated with particle RB motion. The mobilities (K_r and K_t) have been assumed to be isotropic and independent of grain boundary characters. The effect of GB microstates on the defect-GB mediated particle RB motions and sintering kinetics are thus ignored in the current study.

5. Conclusion

A multi-phase-field model has been developed to investigate the interplay between coarsening and densification in solid-state sintering that considers simultaneously mass transport along different paths (including surface diffusion, grain boundary (GB) diffusion, bulk diffusion, vapor transport) and microstructure evolution (including neck growth, pore rounding and shrinkage) as well as the cooperative rigid body (RB) motions (translation and rotation) of particles as they fuse. The simulation results have shown clearly the entire process leading to the complete closure of the internal pore in a three-particle green body, and demonstrated the couplings among diffusion along different paths, neck growth with attendant increase of GB area, pore rounding and shrinkage, and particle RB motions involved in different stages of the process. The relative contribution of each of the processes changes as the sintering progresses and the interplay among them depends on both thermodynamic and kinetic factors of the interfaces (surface and grain boundaries) as well as the particle size. Detailed findings include:

1. The shrinkage of the internal pore consists of three consecutive stages: initial stage for neck formation, intermediate stage for neck growth (increase of GB area) and pore rounding, and the final stage for pore closure. Surface transport and GB transport dominate in the initial and intermediate stages, respectively.
2. Particle RB translation promotes both neck growth and pore shrinkage by transporting mass from the grain interior to GB and pore regions.
3. Particle RB translation forces dynamically evolve as the particles fuse and dominate at the initial and early intermediate stages of neck growth.
4. The force acting on an individual particle varies with not only its size, but also the number and sizes of the neighboring particles.
5. Particle RB rotation does not contribute to either neck growth or the shrinkage of the internal pore among spherical particles, since the associated advective mass flux enters and leaves the pore

region with the same magnitude. This finding stands irrespective of particle size distribution, at least for spherical particles.

6. Three factors are identified that contribute to enhanced sintering kinetics due to reduced particle size: initial pore size (area), diffusion flux, and advective flux due to particle RB translation. The first and last factors are inversely proportional to the square and cube of the particle size, respectively.
7. Surface transport (e.g., surface diffusion) and particle RB motion cooperatively promote neck growth. The former initiates neck formation and growth, which in turn leads the latter to become operative, thereby enhancing further neck growth and preserving the driving force for sintering.

Although most simulations have been carried out for a green body that consists of three spherical particles, the findings above are not expected to change qualitatively in a realistic green body with thousands of particles with a broad size distribution. The findings greatly advance our fundamental understanding of the interplay between the densification and coarsening processes involved in solid-state sintering and may find immediate application in the optimization of green body characteristics to improve sintering kinetics and microstructure.

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